

1. PROJECT & TEAM INFORMATION

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models
- **Team ID:** SWTID-2026-7649
- **Team Leader:** N.Sopya

2. BACKEND ARCHITECTURE

- **Framework:** FastAPI (Python 3.10+) - Chosen for its exceedingly high performance, asynchronous capabilities, and automatic API documentation rendering (/docs).
- **Server:** Uvicorn - A lightning-fast ASGI web server implementation for Python, used to serve the FastAPI application locally.
- **Environment Management:** python-dotenv utilized for managing sensitive API keys and environment variables securely.

3. FRONTEND ARCHITECTURE

- **Templating Engine:** Jinja2 - Integrated seamlessly with FastAPI to dynamically render HTML pages with backend data, allowing for loops and conditional rendering of comic panels.
- **Markup & Styling:** HTML5, CSS3 - Utilized for building a highly responsive, aesthetic, and user-friendly interface without relying on heavy frontend frameworks.

4. ARTIFICIAL INTELLIGENCE INTEGRATION

- **LLM (Large Language Models):**
 - **Google Gemini 1.5 Flash:** Utilized for rapid, structured JSON outline generation based on strict prompt constraints.
 - **Google Gemini 1.5 Pro:** Utilized for creative, detailed narrative, and dialogue generation to breathe life into the story.
- **Diffusion Model (Text-to-Image):**

- **Hugging Face Inference API (Stable Diffusion):** Specifically `runwayml/stable-diffusion-v1-5`, used for rendering highly detailed comic panels based on AI-generated scene descriptions and user art style parameters.

5. UTILITY LIBRARIES

- **Image Processing:** Pillow (PIL) - Used for backend image manipulation, resizing, and drawing the dynamic fallback text-images during API failures.
- **PDF Generation:** FPDF - A lightweight library utilized to compile the final images and narrative text into a structured, downloadable PDF document.
- **HTTP Client:** Python's native `urllib.request` utilized for fetching external APIs efficiently, handling proxies, and bypassing SSL verification constraints.